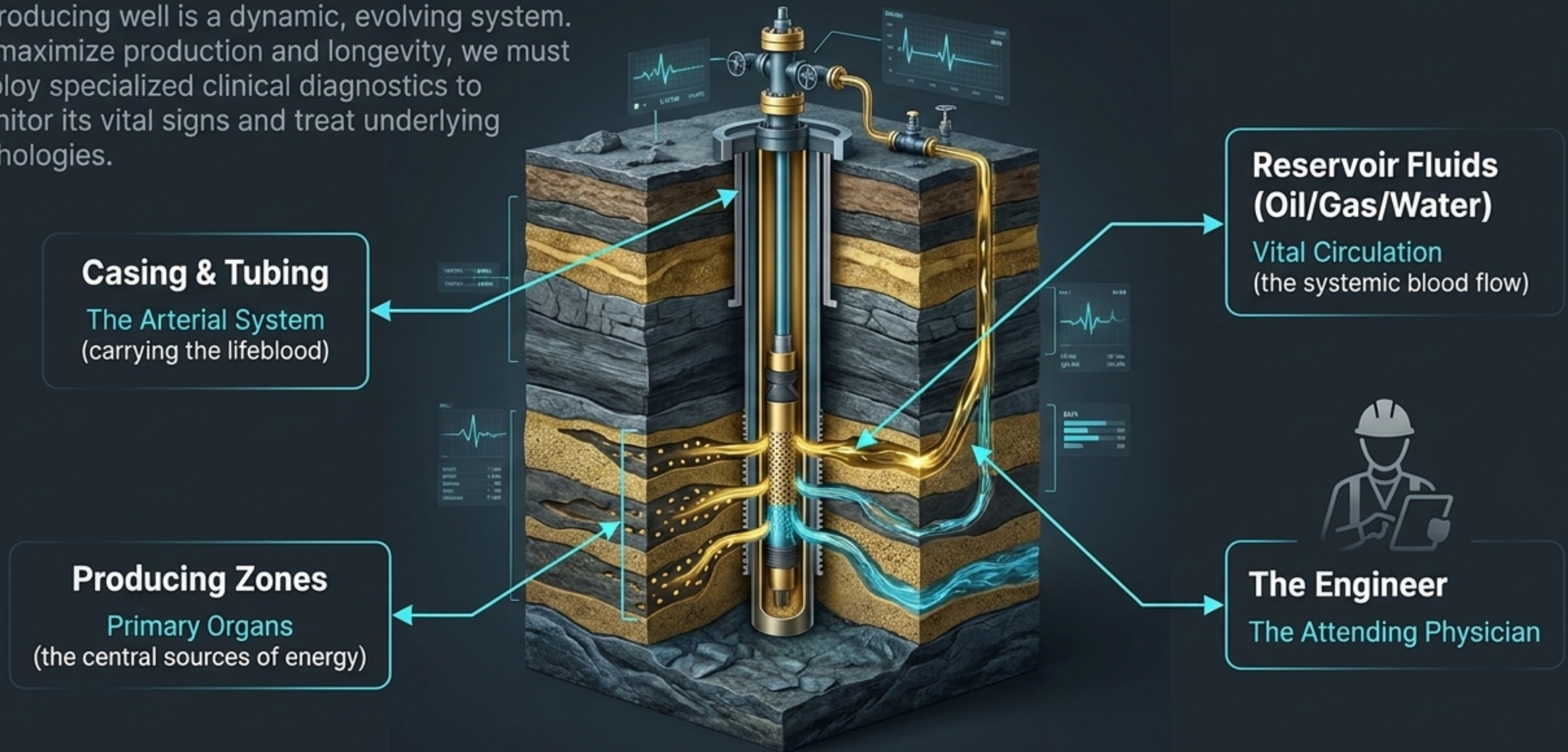


The Well as a Living Patient

A producing well is a dynamic, evolving system. To maximize production and longevity, we must deploy specialized clinical diagnostics to monitor its vital signs and treat underlying pathologies.



Reading the Vitals: The Big Four



THE PULSE (Flow Rate)

Metric: Fluid Velocity

Instrument: Spinner Flowmeter

Diagnostic Value: Measures the speed and rhythm of production.

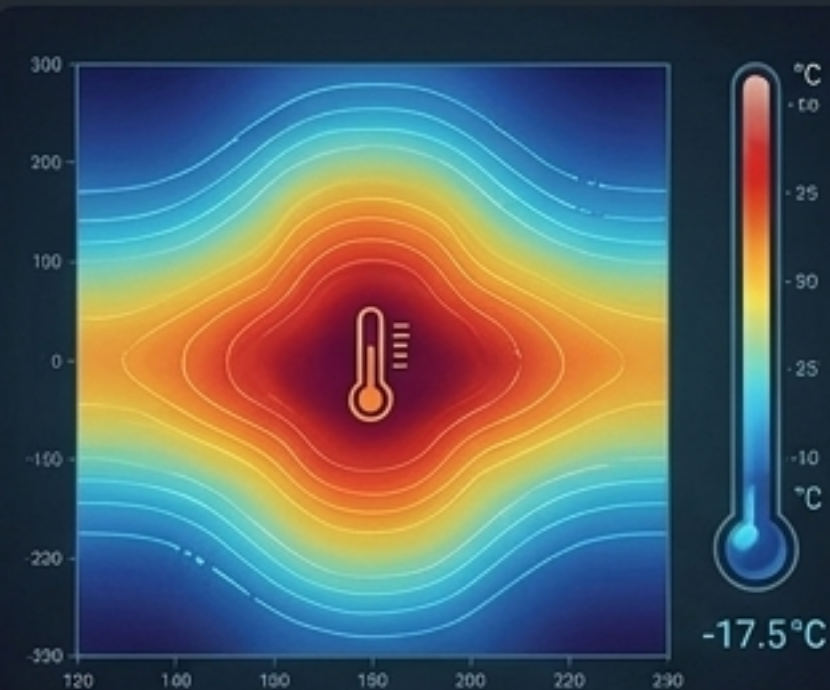


BLOOD PRESSURE (Downhole Pressure)

Metric: Drive Energy & Formation Pressure

Instrument: Pressure Gauge

Diagnostic Value: Indicates the reservoir's underlying strength to push fluids to the surface.

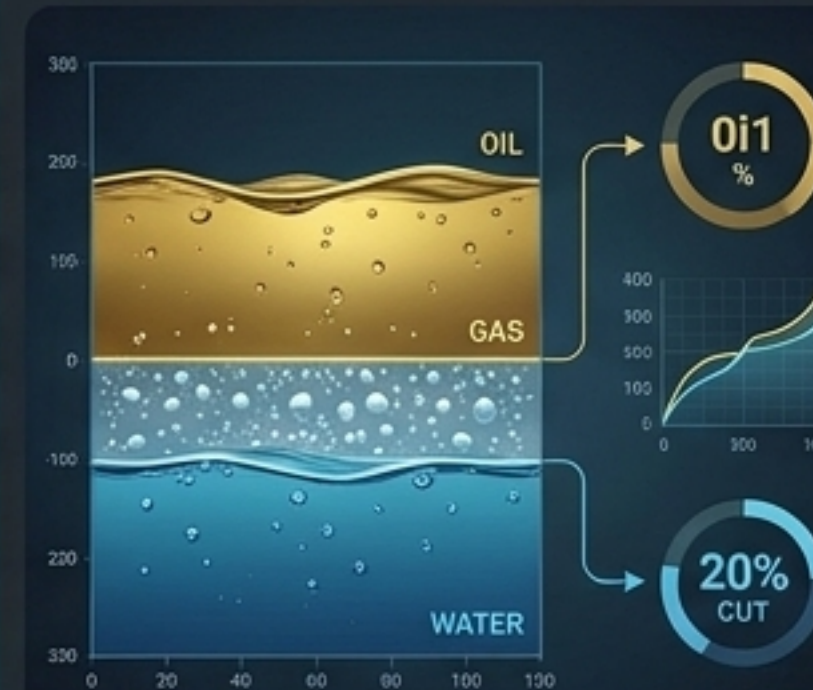


TEMPERATURE (Thermal Output)

Metric: Heat Signatures

Instrument: Temperature Sensor

Diagnostic Value: Detects hidden fluid entry or exit points (the well's fever).



BLOOD COMPOSITION (Holdup / Water Cut)

Metric: Oil / Gas / Water Proportions

Instrument: Fluid Density & Capacitance Tools

Diagnostic Value: Identifies fatal infections like severe water breakthrough.

Translating the Symptoms: What the Earth is Telling Us

	Data Symptom	Diagnostic Translation
	Sudden, localized temperature spike	A new fluid entry point or gas expansion (The well has localized inflammation)
	Acoustic noise detected behind the casing	Uncontrolled channeling, leaks, and fluid movement outside the pipe (Internal bleeding)
	Sudden increase in fluid density / water holdup	Severe water breakthrough into the producing zone (Systemic infection requiring immediate intervention)

Prepared by Eng. Sajjad k. Hano

The Non-Invasive Toolkit: Production Logging Tools (PLTs)

The Stethoscope (Auditory Analysis)

Tool: Noise Log.
Function: Listens for leaks and channeling behind casing.



The Ultrasound & X-Ray (Imaging & Depth)

Tools: Caliper Tool & Gamma Ray Tool.
Function: Maps wellbore diameter, detects casing damage, and correlates depth to identify precise producing intervals.



The MRI (Internal Fluid Mapping)

Tools: Capacitance Tool & Fluid Density Tool.
Function: Sees through the fluid to measure exact water cut, fluid distribution, and holdup without stopping the flow.



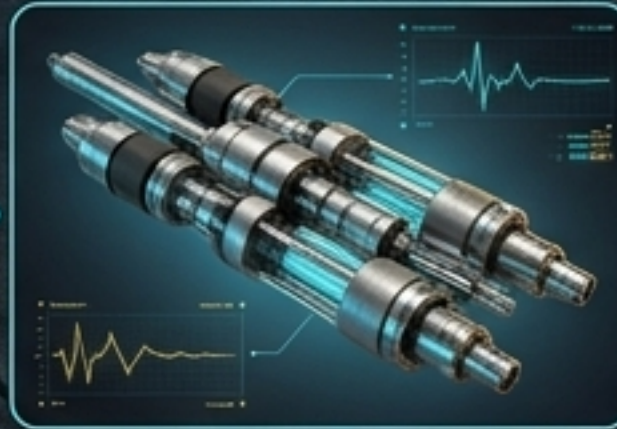
The Deep Biopsy: Formation & Well Testing Tools



RFT (Repeat Formation Tester) - The Targeted Biopsy

Detail: Measures formation pressure and collects fluid samples from individual zones precisely without significant formation damage.

Depth: 1200 ft | Pressure: 1500 psi



MDT (Modular Formation Dynamics Tester) - The Advanced Dynamic Screen

Detail: Executes highly accurate formation pressure testing and pristine fluid sampling for complex diagnostics.

Depth: 3500 ft | Pressure: 3200 psi | Fluid Type: Oil/Water

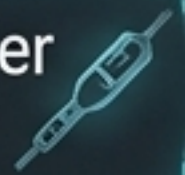


DST (Drill Stem Test) - The Pre-Op Evaluation

Detail: Evaluates reservoir productivity, pressure, permeability, and fluid characteristics before final completion.

Depth: 5800 ft | Flow Rate: 2500 bpd | Permeability: 250 mD

Diagnostic Matrix: Matching Tool to Pathology

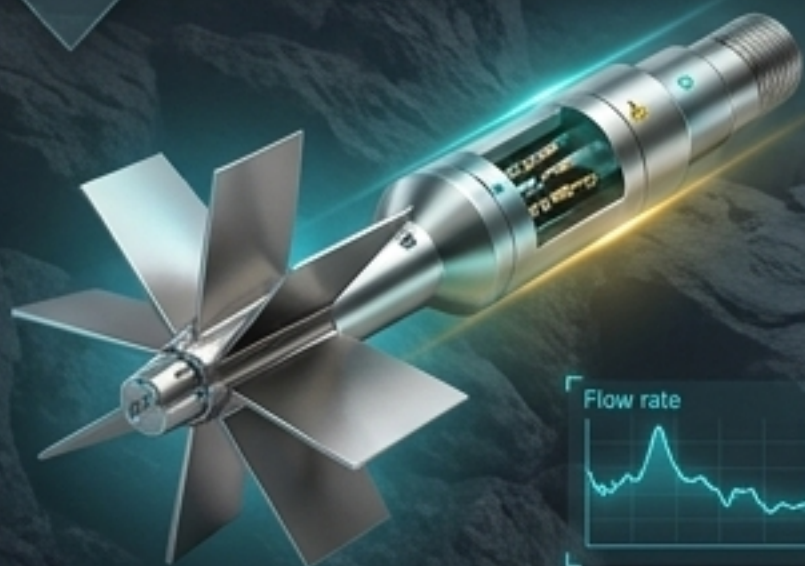
Well Condition (Pathology)	Primary Diagnostic Tool	Expected Data Output
Unknown Producing Zones	Gamma Ray & Spinner Flowmeter 	Depth correlation combined with fluid velocity.
Suspected Casing Damage	Caliper Tool & Noise Log 	Borehole diameter changes and acoustic leak detection.
Formation Pressure Depletion	MDT or RFT 	High-accuracy localized zone pressure.
Severe Water Cut	Capacitance / Water Holdup Tool 	Exact proportion and distribution of oil vs. water.

The Clinical Consultation: Interpreting the Data



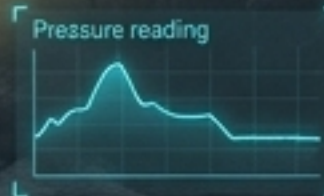
Dr. Flow (Production Engineer)

The Spinner Flowmeter is king. If fluid velocity is zero at the surface, all the pressure in the world doesn't matter. Flow is life.



Dr. Pressure (Reservoir Engineer)

Nonsense. The MDT gives the underlying pressure. Flow is just a symptom; formation pressure is the actual heartbeat of the reservoir. Without drive, the well dies.



Dr. Holdup (Petrophysicist)

You're both blind if you don't know WHAT is flowing. The Capacitance tool reveals the water cut. Pumping massive volumes at high pressure is useless if it's all water. Composition is everything.





MEASURE. ANALYZE. OPTIMIZE. MAXIMIZE.

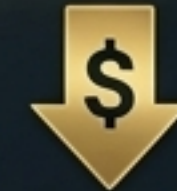
The right downhole data leads to the right production decisions—
and every successful well starts with accurate measurements.



Better Wells



Higher Recovery



Lower Costs

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